

**Statistics 6910:
Introduction to R
Fall 2013**

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Based on Materials provided by Deb Nolan (UC Berkeley) and Duncan Temple Lang (UC Davis).

Theme

- Use the computer expressively to conduct statistical analysis of data.
- Use existing software rather than build routines from the ground up.

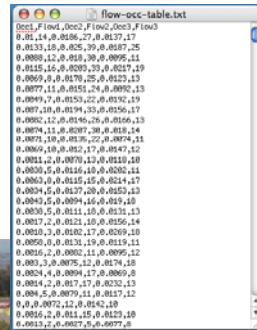
Theme

- Statistical Thinking in the context of computing with data
- DATA Technologies – Statisticians' work includes interfacing and working closely with the original data and those who own it
- Get prepared to handle "Big Data", i.e., data sets that are so large and complex that cannot be handled by conventional tools and methods (think of distributed storage and parallel computing)

What Are Data?

Numbers

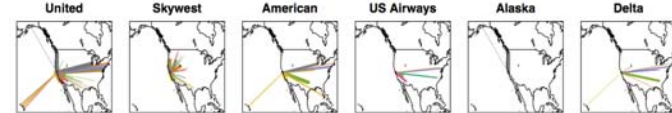
- Example: Traffic on I-80



Dates, Times, Locations

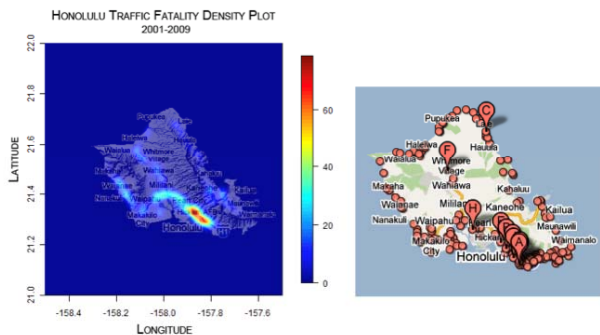
- Example: Flight information

Year	Month	DayOfMonth	DayOfWeek	DepTime	CRSDepTime	ArrTime	CRSArrTime	
1	2007	1	2	1851	1825	1401	1340	
2	2007	1	2	1958	1935	2255	2245	
3	2007	1	2	742	735	1847	1858	
4	2007	1	2	1122	1055	1735	1705	
5	2007	1	2	1142	1105	1400	1335	
6	2007	1	2	2024	2005	2242	2235	
UniqueCarrier FlightNum TailNum ActualElapsedTime CRSElapsedTime AirTime ArrDelay								
1	WN	1719	N3821W	138	135	121	21	
2	WN	1896	N464	125	130	112	10	
3	WN	2296	N462	125	135	116	-3	
4	WN	2459	N405	253	250	239	30	
5	WN	622	N6321W	78	90	69	25	
6	WN	1752	N455	78	90	70	7	
DepDelay Origin Dest Distance TaxiIn TaxiOut Cancelled CancellationCode Diversed								
1	26	OAK	ABQ	889	3	6	0	0
2	15	OAK	ABQ	889	7	6	0	0
3	7	OAK	ABQ	889	3	6	0	0
4	27	OAK	DNA	1959	5	9	0	0
5	37	OAK	BOI	511	3	6	0	0
6	19	OAK	BOI	511	3	5	0	0
CarrierDelay WeatherDelay NASDelay SecurityDelay LateAircraftDelay								
1	21	0	0	0	0	0	0	
2	0	0	0	0	0	0	0	
3	0	0	0	0	0	0	0	
4	21	0	3	0	0	6	0	
5	4	0	0	0	0	21	0	
6	0	0	0	0	0	0	0	



Dates, Times, Locations

- Example: Traffic fatalities



Text

- Example: SPAM or HAM?

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Statistical Thinking and the Data Analysis Cycle

- **Data ACQUISITION** – Input/output, regular expressions
- **Data CLEANING** – Verification, manipulation
- **Data ORGANIZATION** – Data frames, data bases, XML
- **Data ANALYSIS** – Fit and assess statistical models, conduct exploratory data analysis
- **Data SIMULATION** – Simulation studies to understand behavior of data
- **Data REPORTING** – Report findings

Statistical Concepts

- Basic numeracy
 - Variability, Patterns, comparisons
- Graphics
 - Elements and principles of graphing
- Computationally intensive methods, e.g.,
 - Classification and Regression trees, multi-dimensional scaling
- Simulation tools
 - Monte Carlo, bootstrap, cross-validation

Computing Concepts

- Programming concepts
 - Control flow, trees, recursion
- Regular expressions and text manipulation
- Relational databases
- Random number generation
- Representation of information in the computer
- Event handling and GUI development

Software

- R – statistical software
- SQL – structured query language for relational databases
- XML – Extensible Markup Language (and HTML)
- JavaScript – scripting language implemented in most browsers